

Lecture 8:

Text Analysis I: Supervised

IS 618: Social Media Data Analysis
Indira Sen, FSS 2026

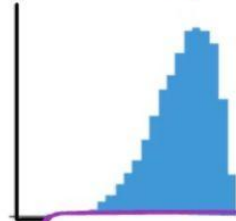
Recap: Data Exploration and Visualization

Exploratory Data Analysis

- Summary and descriptive statistics (mean, median, mode, quantiles, min, max)
- Computing distributions of various variables
- Creating new variables from existing ones
 - Length of content
 - # entities mentioned in the content (using Named Entity Recognition)
 - # links in the content (using regex)
 - (and their domains...)
- Bivariate interactions, e.g.,:
 - Length of content across different user groups
 - Difference in content length by user groups (verified vs. unverified)
-

EDA

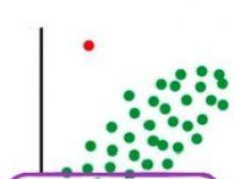
Exploratory Data Analysis



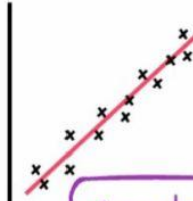
Data distribution

	F	G	H	I	J
A	0.620576	0.140053	1.352728	NaN	0.808078
B	NaN	0.526829	NaN	NaN	0.170902
C	NaN	0.458827	1.406713	0.071119	NaN
D	NaN	2.307197	NaN	NaN	NaN
E	0.203402	0.259913	NaN	0.505811	1.516755

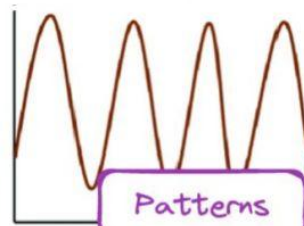
Missing data



Outliers



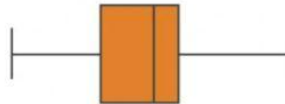
Correlation



Patterns

```
Cust_No      int64
Cust_Name    object
Product_id   int64
Product_cost float64
Purchase_Date datetime64[ns]
dtype: object
```

Data types



Data visualization

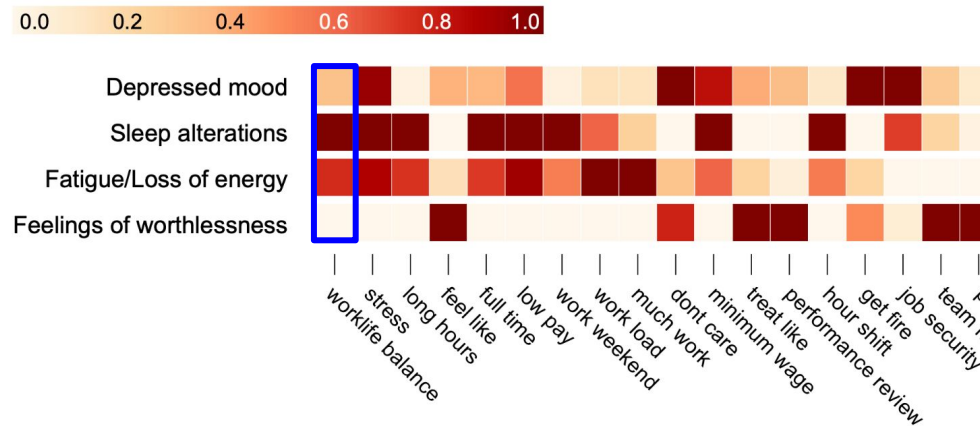


Data quality

<https://www.markovm.com/blog/exploratory-data-analysis>

What TF-IDF tell us about our data

- After calculating the TF-IDF of terms
 - rank them to find important terms
 - across different types of documents
 - one set of tweets could be considered a single document, e.g., tweets from verified users vs. 'regular' users
- Can extend to multi-words: **n-grams** (e.g., **bigrams**, **trigrams**) to capture phrases or collocations.
- TF-IDF can also be used as features for document classification (more in lecture 8)



Sen, Indira, et al. "[Depression at work: exploring depression in major US companies from online reviews](#)." *Proceedings of the ACM on Human-Computer Interaction* 6.CSCW2 (2022): 1-21.

General EDA for Social Media Data

- **Temporal patterns:** post frequency, time-of-day distributions, viral spread timelines
- **Engagement metrics:** likes, shares, comments, views as a function of media type/content
- **Geolocation:** EXIF data or tagged locations for spatial clustering
- **Platform-specific signals:** hashtags, captions, alt-text, account follower counts

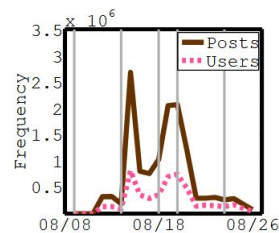
De Choudhury, Munmun, et al. "[Social media participation in an activist movement for racial equality](#)." *Proceedings of the international aaai conference on web and social media*. Vol. 10. No. 1. 2016.

Social Media Participation in an Activist Movement for Racial Equality

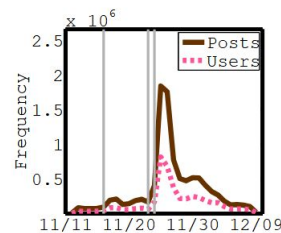
Munmun De Choudhury[†], Shagun Jhaver[†], Benjamin Sugar[†], Ingmar Weber[§]

[†] Georgia Institute of Technology, [§] Qatar Computing Research Institute, HBKU
{munmund, jhaver.shagun, bsugar}@gatech.edu, iweber@qf.org.qa

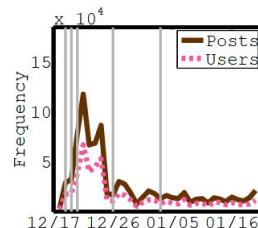
From the Arab Spring to Black Lives Matter, social media has been instrumental in the rise of political movements throughout the world. We present one of the first systematic studies of social media participation in an activist movement around Black Lives Matter, known as "Black Lives Matter." We use social media as a sensor for the events related to the movement over time, the geography, and its relationship to protest events. We find evidence for temporally separated events related to the movement. We find that participants from regions of black victimization due to greater negativity and make more negative comments. Finally, we observe that social media participation and language can predict protest events on the ground. We discuss the role of social media in collective action around this movement and how social media platforms may help in socially contested and sensitive



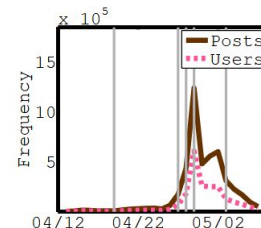
(a) Ferguson I



(b) Ferguson II



(c) NYC



(d) Baltimore

Tweet and user volume over time and geography about "Black Lives Matter"

Data Visualization

General Data Visualization

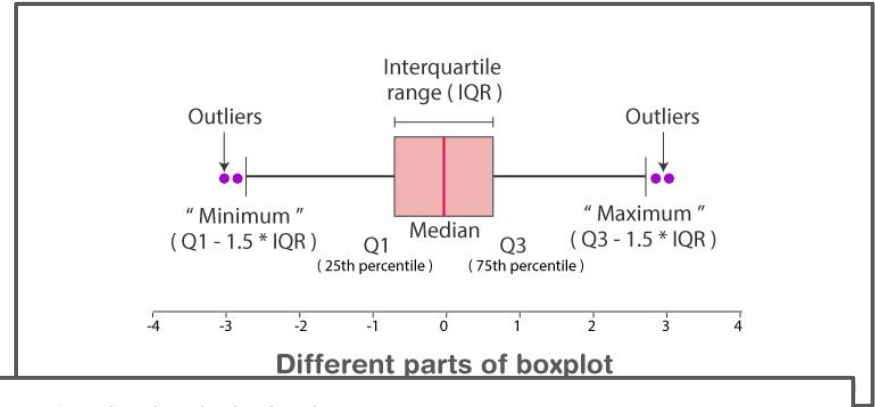
Histograms

Bar Charts

Box Plots

Pie Charts

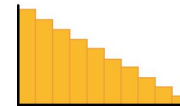
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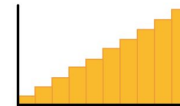
Symmetric (normal) vs skewed and uniform distributions



Normal distribution
(unimodal, symmetric,
the "bell curve")



**Right-skewed
distribution**
(Positively-skewed)



**Left-skewed
distribution**
(Negatively-skewed)



Uniform distribution
(equal spread,
no peaks)

Unimodal vs bimodal distributions



Normal distribution
(unimodal, symmetric,
the "bell curve")



**Symmetric bimodal
distribution**
(two modes)



**Non-symmetric
bimodal distribution**
(two modes)

Examples of Data Visualization in Social Media Research: Mental Health on Reddit

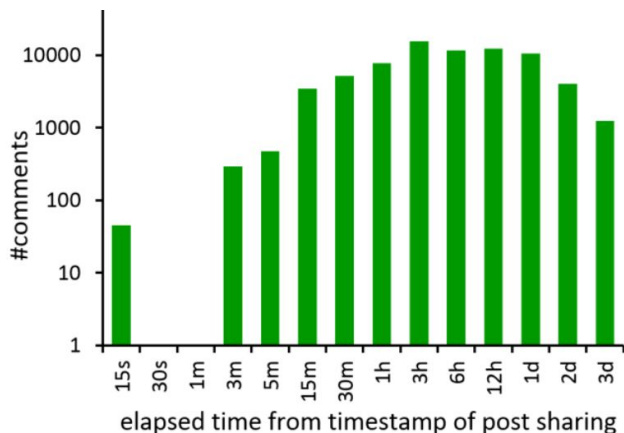


Figure 2. Comment distribution over time. Each bin shows the number of comments shared within a certain time interval from the timestamp of post.

De Choudhury, Munmun, and Sushovan De. "[Mental health discourse on reddit: Self-disclosure, social support, and anonymity.](#)" *Proceedings of the international AAAI conference on web and social media*. Vol. 8. No. 1. 2014.

Mental Health Discourse on reddit: Self-Disclosure, Social Support, and Anonymity

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Abstract

Social media is continually emerging as a platform of information exchange around health challenges. We study mental health discourse on the popular social media: reddit. Building on findings about health information seeking and sharing practices in online forums, and social media like Twitter, we address three research challenges. First, we present a characterization of self-disclosure in mental illness communities on reddit. We observe individuals discussing a variety of concerns ranging from the daily grind to specific

online fora, these social systems are more holistic, in the sense that millions of people use them to post about the mundane going on of their lives. Additionally, unlike many online communities, most social media sites have a personal permanent identity associated with user profiles. Consequently, these platforms provide a rich ecosystem to study the variety of self-disclosure, social support, and disinhibition that engender health related discourse.

Specifically, we examine a highly popular social news and entertainment media: reddit (<http://www.reddit.com/>).

n interesting online social system that has the if a forum: it allows sharing blurbs of text and posts that invite votes and commentary. At the it is often used as a social feed of information l from people's contacts and audiences. Another aspect of reddit is its "throwaway" accounts. emporary identities often used as an "anonymity scuss uninhibited feelings, sensitive information, unacceptable thoughts momentarily; information considered unsuitable for the mainstream. e unique characteristics of reddit, in this paper rested in studying the nature of discourse around ant health challenge of mental illness. Mental

Posts per day	247.1 (± 41.3)
Average posts per user	0.75 (± 2.21)
Mean, Median length of posts (words)	96.4, 64
Average upvotes per post	8.37 (± 23.2)
Average downvotes per post	1.26 (± 3.71)
Average comments per post	5.03 (± 7.52)
Mean, Median comment length (words)	27.03, 16
Average comments per user	2.69 (± 8.49)
Average upvotes per comment	2.39 (± 4.16)
Average downvotes per comment	2.06 (± 3.58)
Total number of posts	72,236

Table 1. Basic descriptive statistics of the reddit dataset.

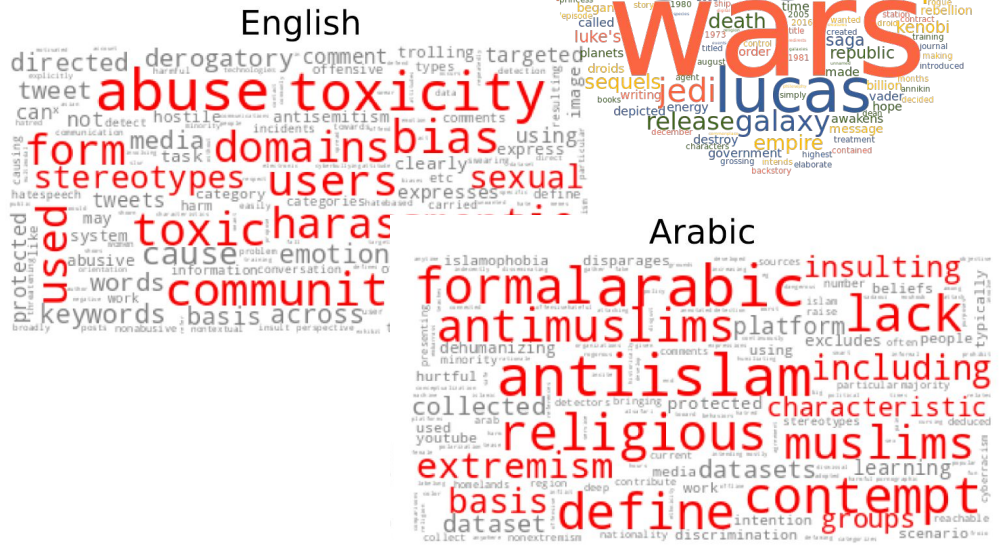
Data Visualization for Text

- Word clouds: easy to make, but often too vague.



Data Visualization for Text

- Word clouds: easy to make, but often too vague.
- Variants of word clouds
 - Accentuated word clouds



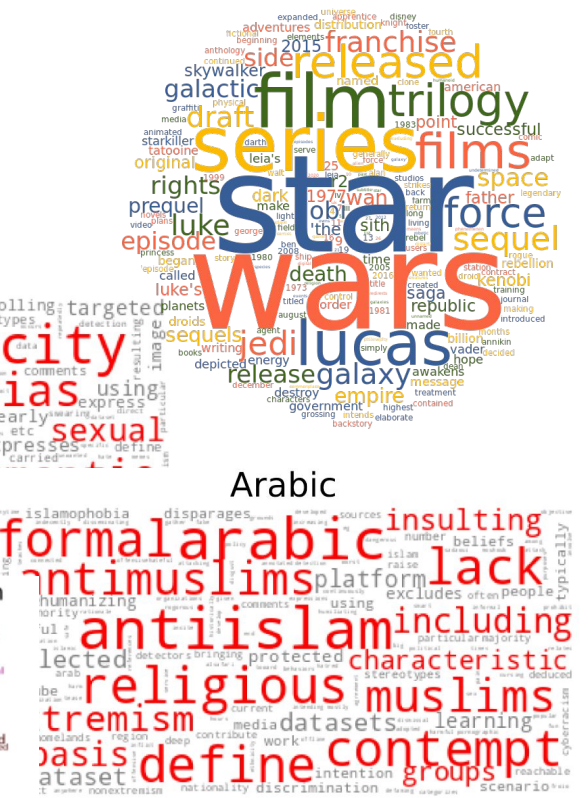
Data Visualization for Text

- Word clouds: easy to make, but often too vague.
- Variants of word clouds
 - Accentuated word clouds
 - Semantically grouped word clouds



Data Visualization for Text

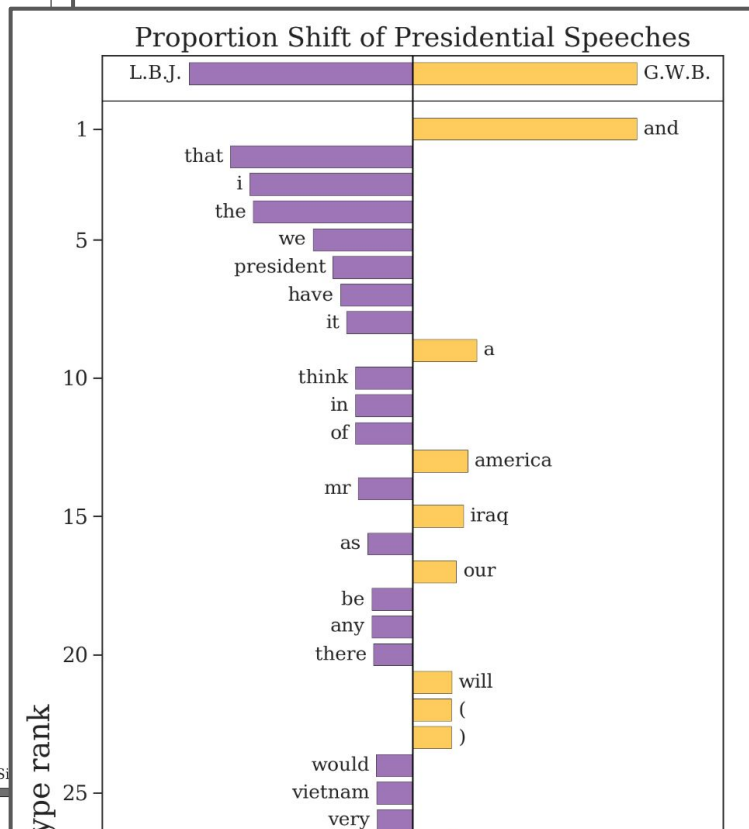
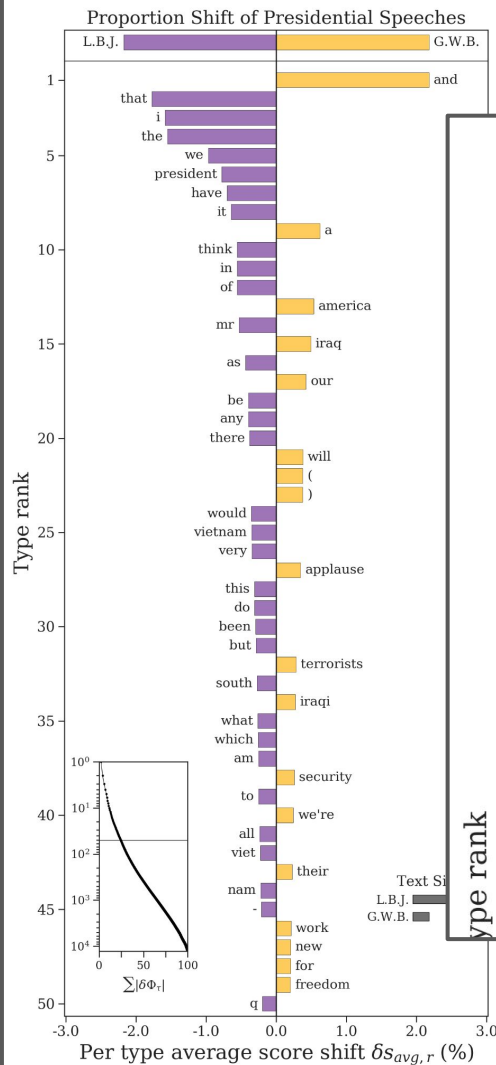
- Word clouds: easy to make, but often too vague.
- Variants of word clouds
 - Accentuated word clouds
 - Semantically grouped word clouds
- Word Shift Graphs



Word Shift Graphs

- Used for visualizing **pairwise comparisons** between texts through word shifts
- Word shift graphs extract which words contribute to a difference between two texts

<https://shifterator.readthedocs.io/en/latest/>



Using Word Shift Graphs to Study Information Manipulation

Coordinated Behavior on Social Media in 2019 UK General Election

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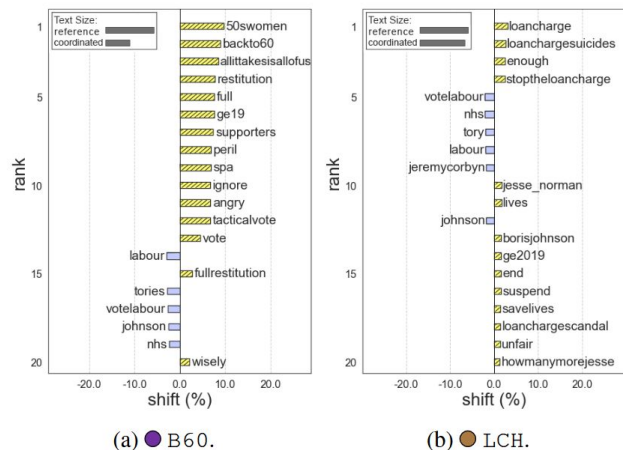


Figure 8: Application of word shift graphs for highlighting narratives that characterize coordinated communities. Words are ranked based on their contribution towards differentiating coordinated and non-coordinated users. Yellow-colored words (right-hand side of each word shift graph) are informative for coordinated users, while blue-colored words (left-hand side) are informative for non-coordinated users.

Abstract

Coordinated online behaviors are an essential part of information and influence operations, as they allow a more effective disinformation's spread. Most studies on coordinated behaviors involved manual investigations, and the few existing computational approaches make bold assumptions or oversimplify the problem to make it tractable.

Here, we propose a new network-based framework for uncovering and studying coordinated behaviors on social media. Our research extends existing systems and goes beyond limiting binary classifications of coordinated and uncoordinated behaviors. It allows to expose different coordination patterns and to estimate the degree of coordination that characterizes diverse communities. We apply our framework to a dataset collected during the 2019 UK General Election, detecting and characterizing coordinated communities that participated in the electoral debate. Our work conveys both theoretical and practical implications and provides more nuanced and fine-grained results for studying online information manipulation.

Nizzoli, Leonardo, et al. "[Coordinated behavior on social media in 2019 UK general election.](#)" Proceedings of the international AAAI conference on web and social media. Vol. 15. 2021.

Alassad, Spann, amos 2017). IOs can take different shapes, target various individuals, online crowds, or communities, and have diverse goals (Starbird, Arif, and Wilson 2019). Among the strategic tools used by propagators are fake news, propaganda, hateful speech, or

efforts, researchers debate the efficacy of the proposed solutions, and IOs still appear to pose a severe threat to our democracies and societies (Barrett 2019).

Meanwhile, groundbreaking advances in specific areas of computing are causing profound changes to the online information landscape. Advances in artificial intelligence brought to the rise of deepfakes – synthetic media where the source has been modified via deep learning techniques. Deepfakes allow crafting arbitrary texts that resemble a target person's writing style or audio and video samples where a target person's face and voice could be made to do or say anything. Unsurprisingly, these powerful techniques have already been used to create fake news (Zellers et al. 2019) and fake profile pictures for fraudulent accounts¹. However, each IO must spread to "infect" many users in order to be successful, independently of its aims and tools used to deceive. This goal often mandates extensive and coordinated social media efforts for the campaign to obtain a significant outreach, exert influence, and thus have an impact. In light of this consideration, since 2018, all leading platforms showed great interest in studying *coordinated inauthentic behavior* (CIB)². Despite often appearing together, coordination and inauthenticity are two distinct concepts. For example, activists and other grassroots initiatives typically feature coordinated but uncoordinated behavior. Conversely, a single fake account may have the intent to mislead but exhibit inauthentic but uncoordinated behavior. The majority of existing efforts for studying CIB involved plenty of manual investigations, while the computational approaches are still few and far between. Among the challenges, there are the ambiguity and fuzziness of CIB itself: What exactly is a coordinated behavior? What is inauthentic behavior? How many organized

Related Approaches

- TF-IDF is not the only approach for highlighting significant words
- Word Shift graphs use different options for computing text comparison, e.g., Shannon entropy, Kullback-Leibler divergence
- Check [Monroe et al., 2008](#) for a nice overview
- Subtle differences in the results of different approaches

Fightin' Words: Lexical Feature Selection and Evaluation for Identifying the Content of Political Conflict

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Entries in the burgeoning "text-as-data" movement are often accompanied by lists or visualizations of how word (or other lexical feature) usage differs across some pair or set of documents. These are intended either to establish some target semantic concept (like the content of partisan frames) to estimate word-specific measures that feed forward into another analysis (like locating parties in ideological space) or both. We discuss a variety of techniques for selecting words that capture partisan, or other, differences in political speech and for evaluating the relative importance of these words. We introduce and emphasize several new measures of word importance. Monroe, Burt L., Michael P. Colaresi, and Kevin M. Quinn. "Fightin' words: Lexical feature selection and evaluation for identifying the content of political conflict." *Political Analysis* 16.4 (2008): 372-403.

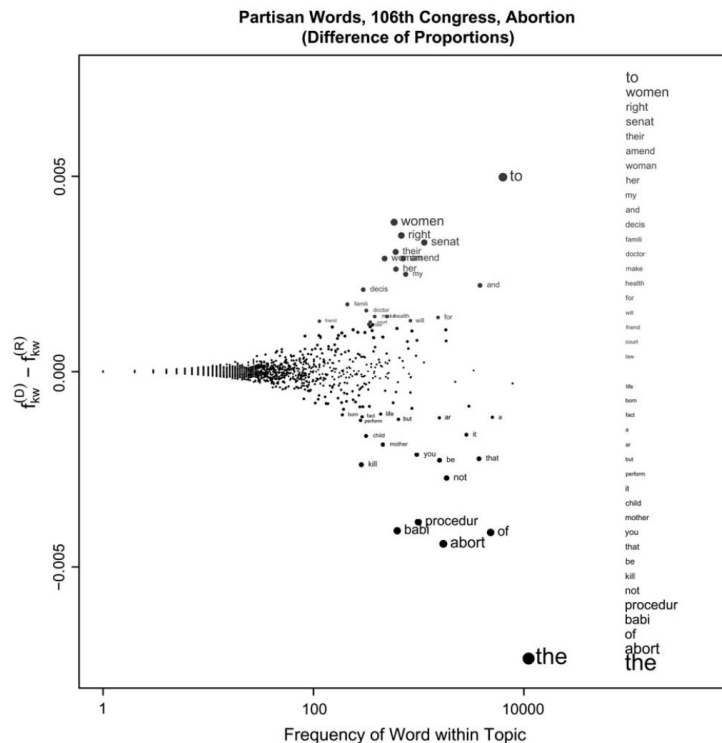


Fig. 1 Feature evaluation and selection using $f_{kw}^{(D)} - f_{kw}^{(R)}$. Plot size is proportional to evaluation weight, $|f_{kw}^{(D)} - f_{kw}^{(R)}|$. The top 20 Democratic and Republican words are labeled and listed in rank order to the right. The results are almost identical for two other measures discussed in the text: unlogged *tf.idf* and frequency-weighted WordScores.

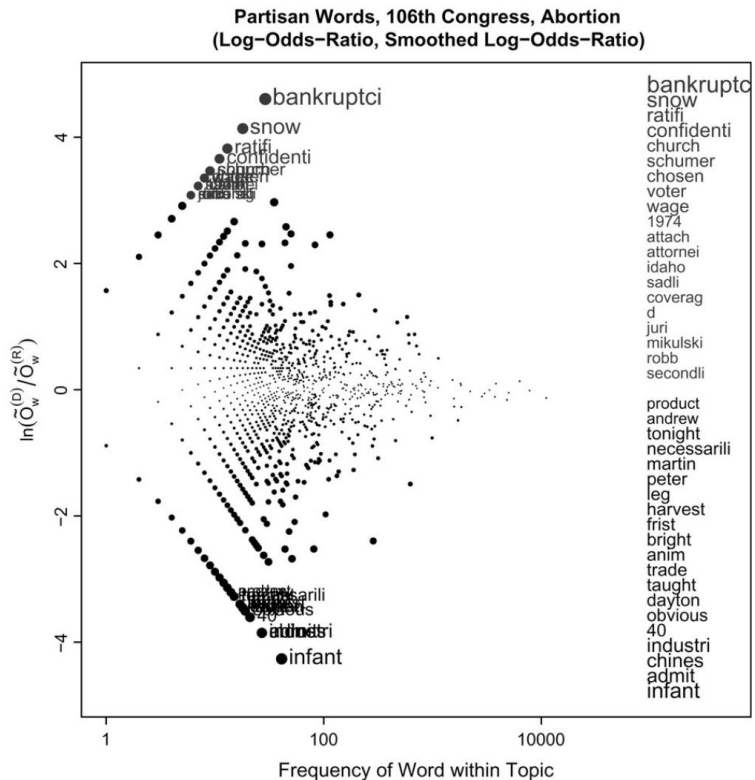


Fig. 2 Feature evaluation and selection using $\hat{\delta}_{kw}^{(D-R)}$. Plot size is proportional to evaluation weight, $|\hat{\delta}_{kw}^{(D-R)}|$. Top 20 Democratic and Republican words are labeled and listed in rank order. The results are identical to another measure discussed in the text: the log-odds-ratio with uninformative Dirichlet prior.

General tips, especially for your reports and presentations

Bakhshi, Saeideh, David A. Shamma, and Eric Gilbert. "[Faces engage us: Photos with faces attract more likes and comments on instagram.](#)" Proceedings of the SIGCHI conference on human factors in computing systems. 2014.

- Descriptive statistics:
 - Tables or figures showing summary of your data
 - especially the variables you are mainly studying (distributions, min, max, averages)

Type	Variable	Description	Distribution
Engagement	likes*	Number of likes on each photo.	
	comments*	Number of comments on each photo.	
Audience & Activity	followers*	Number of users who follow the photo's owner.	
	photos*	Number of photos shared by photo's owner.	
Faces	has face	1 if the photo contains a face, 0 otherwise.	
	has face < 18 years old	1 if there is at least one face younger than 18, 0 otherwise.	
	has face ∈ [18, 35] years old	1 if there is at least one face with age between 18 and 35, 0 otherwise.	
	has face > 35 years old	1 if there is at least one face older than 35, 0 otherwise.	
	has female face	1 if there is at least one female face in the photo, 0 otherwise.	
	has male face	1 if there is at least one male face in the photo, 0 otherwise.	

Table 1. Distributions of quantitative and binary variables used in this paper. Variables marked with “*” are log-transformed. The red and blue lines identify mean and median of the distribution, respectively. Orange refers to 1’s in the bar graphs. The engagement variables are our dependent measures. Audience and activity variables are used as controls, and faces variables are the focal point of this study.

General tips, especially for your reports and presentations

- Data visualization:
 - All axes should have labels
 - Captions should be self-contained
 - The text in the figure should be as big as the text in your paper
 - Using colors to convey information is nice, but also good to use markers or textures for this
 - Also keep in mind accessibility:
 - <https://towardsdatascience.com/two-simple-steps-to-create-colorblind-friendly-data-visualizations-2ed781a167ec>

Today: Text Analysis 1: Supervised Machine Learning

Now that we have the cloned data, what do we do with it?

- Analyze it! Answer your research questions!
- You can use a wide range of **computational** methods, depending on your goal:
 - Text analysis, Natural Language Processing (NLP), also called “text-as-data” methods: analyze textual features of the data — topics, framing, etc
 - Network analysis: e.g., find the most central units of your data
 - Image analysis: use computer vision techniques to find features of images
 - Statistical inference
 - Descriptive statistics: summaries of your data, e.g., mean and variance of your variables
 - Analytic statistics: relationship between variables, studied through regressions, e.g., how are the number of faces in an instagram post associated with the likes it gets (Bakshi et al.,)
 - Causal inference: go beyond correlation, but often hard with observational social media data (more on this later in the course)

Our language, not just from social media, carries a lot of information

- Words reveal **social identity**: dialect, register, and vocabulary choices signal group membership, education, and background
- **Hedging**: “I *suppose* it’s fine” vs. “it’s fine”
- **Temporal change**: words shift in meaning, frequency, and connotation over time, making language a record of cultural change
- ...

Content Analysis: A Systematic Study of Content

- Content analysis:
 - “any technique for making inferences by objectively and systematically identifying specified characteristics of messages” [Holsti, 1969]
 - “a systematic, replicable technique for compressing many words of text into fewer content categories based on explicit rules of coding” [Berelson, 1952; GAO, 1996; Krippendorff, 1980; and Weber, 1990]

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- Questions content analysis can help us answer:
 - How does news media represent the immigration crisis?
 - What are topics that lead to arguments in long-term relationships?
 - How do citizens perceive the performance of politicians during the pandemic?

Content Analysis: A Systematic Study of Content

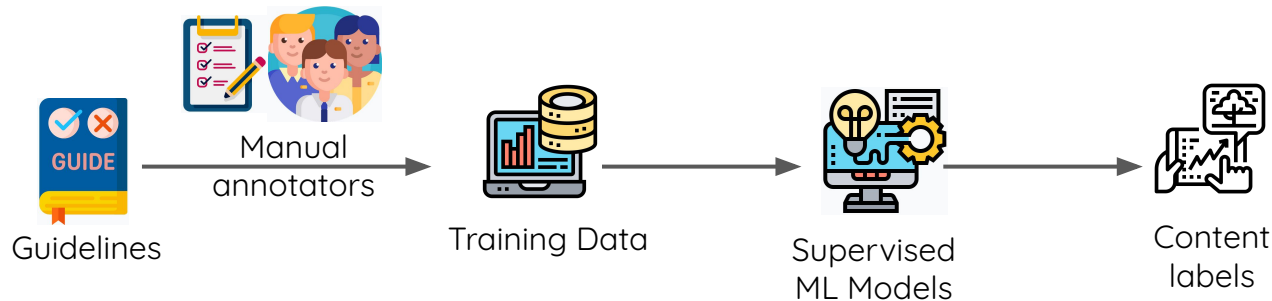
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- Questions content analysis can help us answer:
 - How does news media represent the immigration crisis?
NYtimes articles frames [Mendelsohn’21]
 - What are topics that lead to arguments in long-term relationships?
r/relationshipadvice posts topics
 - How do citizens perceive the performance of politicians during the pandemic?
Twitter posts stance

Content Analysis Pipeline using ML/NLP

- unsupervised



- supervised



Example: Unsupervised

- Kickstarter is a crowdfunding website, a site where artists and entrepreneurs pitch projects and ask people to invest in these pitches
- Mitra and Gilbert study what makes certain pitches more persuasive and more likely to get investment
- Find frequent phrases — unigrams, bigrams, and trigram — that are associated with successful pitches

The Language that Gets People to Give: Phrases that Predict Success on Kickstarter

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ABSTRACT

Crowdfunding sites like Kickstarter—where entrepreneurs and artists look to the internet for funding—have quickly risen to prominence. However, we know very little about the factors driving the “crowd” to take projects to their funding goal. In this paper we explore the factors which lead to successfully funding a crowdfunding project. We study a corpus of 45K crowdfunded projects, analyzing 9M phrases and 59 other variables commonly present on crowdfunding sites. The language used in the project has surprising predictive power—accounting for 58.56% of the variance around successful funding. A closer look at the phrases shows they exhibit general persuasion principles. For example, *also receive two* reflects the principle of *Reciprocity* and is one of the top

Mitra, Tanushree, and Eric Gilbert. “[The language that gets people to give: Phrases that predict success on kickstarter.](#)”

Proceedings of the 17th ACM conference on Computer supported cooperative work & social computing. 2014.

entrepreneurs alike look to the internet for capital. At the time of this writing, a small startup called Pebble¹ is Kickstarter's

to generate press attention [29], yet only attracted one-third of its requested \$10K. As per Kickstarter's “all-or-nothing” funding rules, the Ninja Baseball team received no money.

Successful projects like Pebble have already raised over \$300M on Kickstarter [30]. Furthermore, crowdfunding sites seem poised to rise to greater prominence soon. With President Obama's signing of the JOBS Act and its subsection the CROWDFUND ACT [7], industry observers expect several new crowdfunding sites to emerge very soon [13]. They will join prominent ones already on the internet, like StartSomeGood³ and Flattr⁴, in addition to crowdfunding platforms like Kickstarter, IndiGoGo and RocketHub.

While research in crowdfunding has recently gained attention in the HCI and CSCW communities [17, 21, 27, 35], we still know very little, for example, about what drives the “crowd” to take projects to their funding goals. What makes some projects like Pebble succeed while others, like Ninja Baseball, fail? In this paper, we look to answer this question. Results from our statistical model show that several project attributes (e.g., duration, presence of a video, etc.) have substantial predictive power, yielding a model with fairly high accuracy: 17.03% error under 10-fold cross-validation. We find, however, that the error-rate drops to 2.4% when we include the phrases used in the project's pitch—a non-random improvement over a strong null, controls-only model. This suggests that the language used by creators to pitch their project plays a major role in driving the project's success, accounting for 58.56% of the variance around success.

Adopting a corpus of 45K Kickstarter projects and natural

Example: Supervised

- Framing selects particular aspects of an issue and emphasizes them
 - Impacts how people engage with content
 - Affects public opinion on topics
- Examples of framing immigration
 - as an economic issue
 - as a security issue
- Mendelsohn et al., create a custom classifier that detects different framings of tweets about immigration

Modeling Framing in Immigration Discourse on Social Media

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Abstract

The framing of political issues can influence policy and public opinion. Even though the public plays a key role in creating and spreading frames, little is known about how ordinary people on social media frame political issues. By creating a new dataset of immigration-related tweets labeled for multiple framing typologies from political communication theory, we develop supervised models to detect frames. We demonstrate how users' ideology and region impact framing choices, and how a message's framing influences audience responses. We find that the more commonly-used *issue-generic* frames change

social media content enables us to compare framing strategies across countries and political ideologies. Furthermore, social media provides unique insights into how messages resonate with audiences through interactive signals such as retweets and favorites. By jointly analyzing the production and reception of frames on Twitter, we provide an in-depth analysis of immigration framing by and on the public.

Political communications research has identified numerous typologies of frames, such as *issue-generic policy*, *immigration-specific*, and *narrative*. Each of these frame types can significantly shape the audience's perceptions of an issue (Iyengar, 1991; Chong and Druckman, 2007; Lecheler et al., 2015), but prior NLP work seeking to detect frames in mass media (e.g. Card et al., 2016; Field et al., 2018; Kwak et al., 2020) has largely been limited to a single *issue-generic policy* typology. Multiple dimensions of framing must be considered in order to better understand the structure of immigration discourse and its effect on public opinion and attitudes. We thus create a novel dataset of immigration-related tweets containing labels for each typology to facilitate more nuanced computational analyses of framing.

This work combines political communication theory with NLP to model multiple framing strategies and analyze how the public on Twitter frames immigration. Our contributions are as follows:

Mendelsohn, Julia, Ceren Budak, and David Jurgens. "[Modeling framing in immigration discourse on social media](#)." Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. 2021.

1991), and endorse possible solutions, thus having major implications for public opinion and policy decisions (Chong and Druckman, 2007). While past

Another Example

- Let's say you want to study the 'AI-sloppiness' of some X/Twitter posts
- You don't have a readymade classifier for this
 - AI Sloppiness != AI-generated
- You have a cleaned dataset of X posts
- You need a **custom** classifier for your data and research question

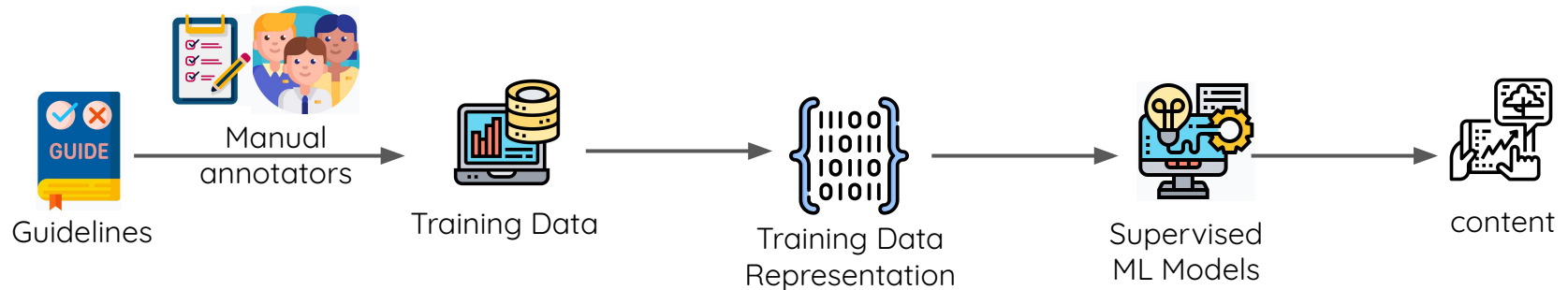
How do we create custom classifiers with our data?

- If we don't have ready-made classifiers for our research questions, then we build our own



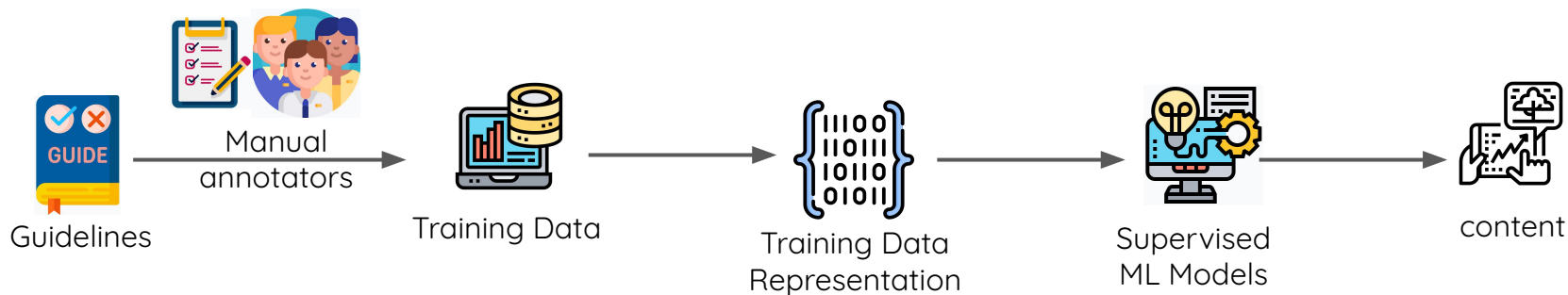
How do we create custom classifiers with our data?

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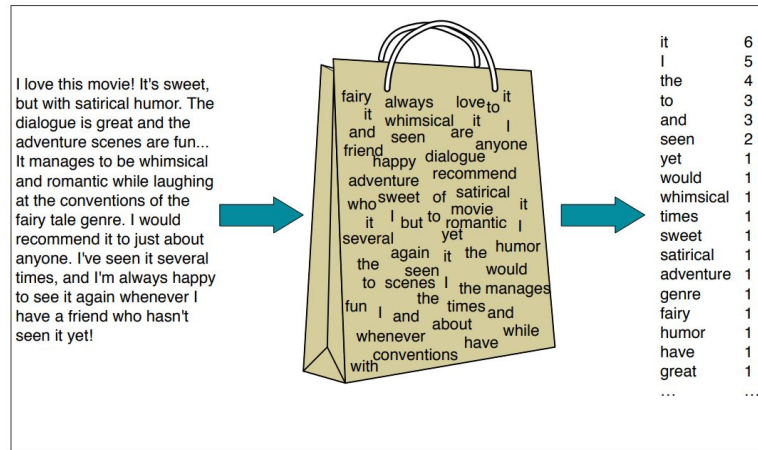
How do we create custom classifiers with our data?

- If we don't have ready-made classifiers for our research questions, then we build our own
- You need a **representation** of your data that a computer or computational model will understand
- Many ways of creating these representations
 - Usually start with tokens
 - Simple representations: TF-IDF
 - Even simpler: 'bag-of-words', or simply term frequency or occurrence
 - More complex: Word and document embeddings (next lecture)



Bag-of-words Representation

- The most basic: Bag-of-words (BoW)
- Bag of Words assumption: word order does not matter



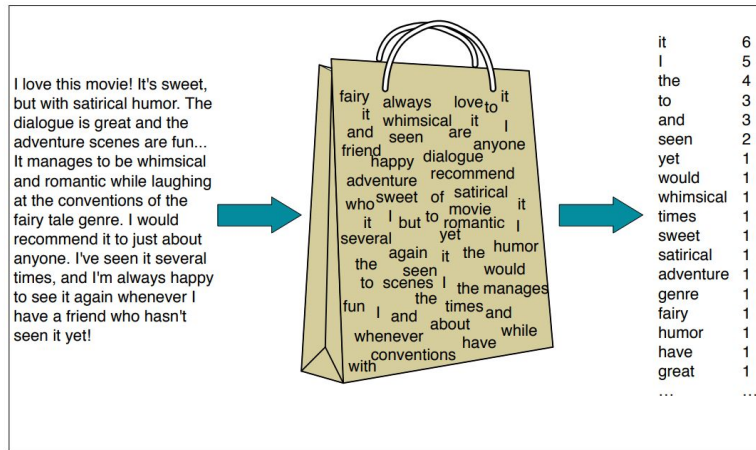
Jurafsky and Martin, Speech and Language Processing

Bag-of-words Representation

- The most basic: Bag-of-words (BoW)
- Bag of Words assumption: word order does not matter



When can this assumption be problematic?



Jurafsky and Martin, Speech and Language Processing

Document D1	<i>The child makes the dog happy</i> the: 2, dog: 1, makes: 1, child: 1, happy: 1
Document D2	<i>The dog makes the child happy</i> the: 2, child: 1, makes: 1, dog: 1, happy: 1



	child	dog	happy	makes	the	BoW Vector representations
D1	1	1	1	1	2	[1,1,1,1,2]
D2	1	1	1	1	2	[1,1,1,1,2]

Document D1	<i>The child makes the dog happy</i> the: 2, dog: 1, makes: 1, child: 1, happy: 1
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a list of numbers: Vector

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a list of numbers: Vector

a list of lists:
Matrix

Let's say we have 4 social media posts in our data

- Documents
 - "I love cheese"
 - "Dogs hate cheese"
 - "I hate chocolate"
 - "I love dogs"
- Convert these to tokens
- Get the total vocabulary
- For each document, create a vector based on whether a token occurs or not

Let's say we have 4 social media posts in our data

- Documents
 - "I love cheese" → ["I", "love", "cheese"]
 - "Dogs hate cheese" → ["dogs", "hate", "cheese"]
 - "I hate chocolate" → ["I", "hate", "chocolate"]
 - "I love dogs" → ["I", "love", "dogs"]
- **Convert these to tokens**
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 - “I love cheese” → [“i”, “love”, “cheese”]
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 - “I hate chocolate” → [“i”, “hate”, “chocolate”]
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- Convert these to tokens
- **Get the total vocabulary** = [“i”, “love”, “cheese”, “dogs”, “hate”, “chocolate”]
(for vocabulary always count unique tokens)
- For each document, create a vector based on whether a token occurs or not

Let's say we have 4 social media posts in our data

- Documents
 - "I love cheese" → ["i", "love", "cheese"] → [1, 1, 1, 0, 0, 0]
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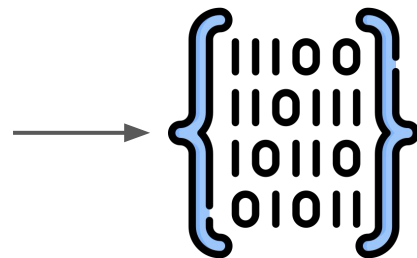
Let's say we have 4 social media posts in our data

- Documents
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 - "Dogs hate cheese" → ["dogs", "hate", "cheese"] → [0, 0, 1, 1, 1, 0]
 - "I hate chocolate" → ["i", "hate", "chocolate"] → [1, 0, 0, 0, 1, 1]
 - "I love dogs" → ["i", "love", "dogs"] → [1, 1, 0, 1, 0, 0]
- Convert these to tokens
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- “I hate chocolate” $\rightarrow$ [“i”, “hate”, “chocolate”] $\rightarrow$ [1, 0, 0, 0, 1, 1]
- “I love dogs” $\rightarrow$ [“i”, “love”, “dogs”] $\rightarrow$ [1, 1, 0, 1, 0, 0]



- Convert these to tokens
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- **For each document, create a vector based on whether a token occurs or not**

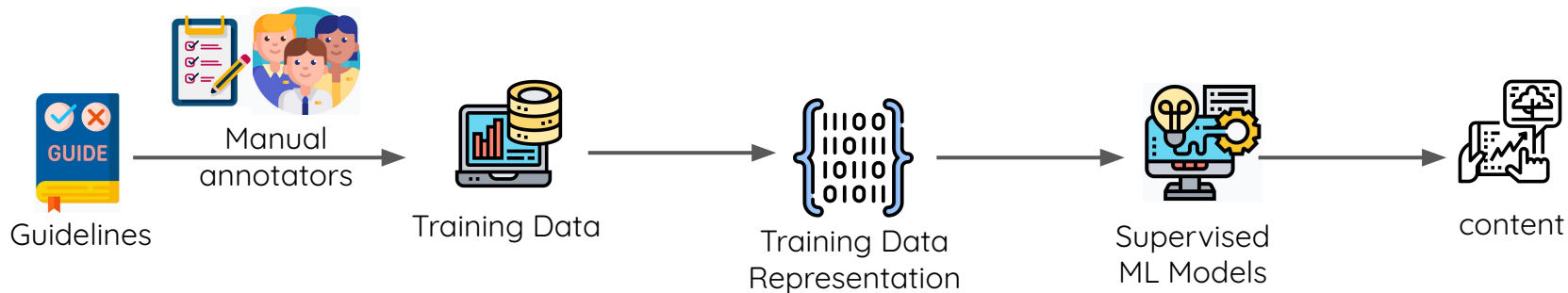
Let's classify sentiment of our documents



How to do supervised classification

We can use the representations we created (e.g., Bag-of-words) to further use for classification.

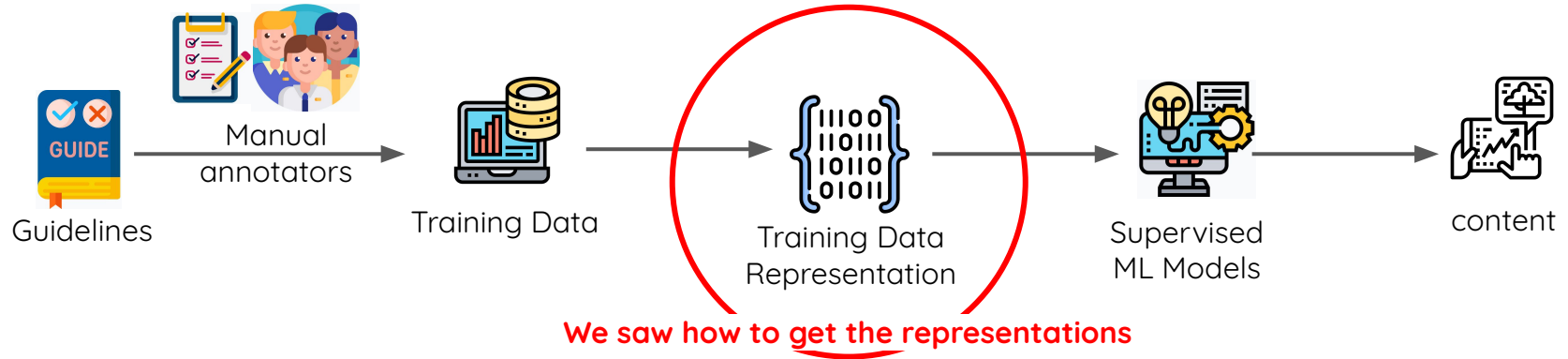
All we need is some labeled data or training data.



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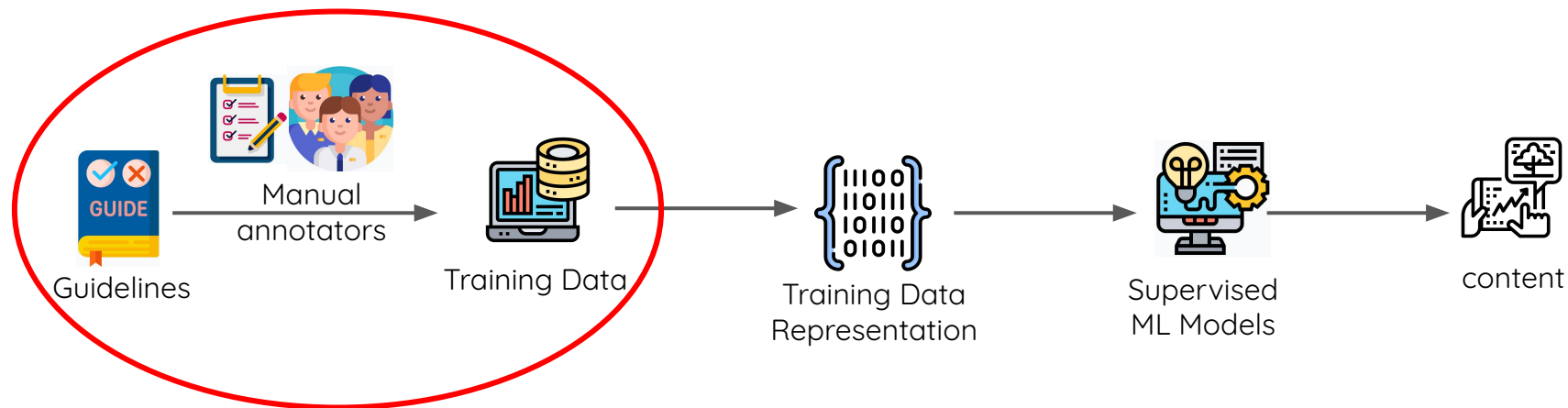
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How to do supervised classification

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Now, all we need is some **labeled data** or **training data**.

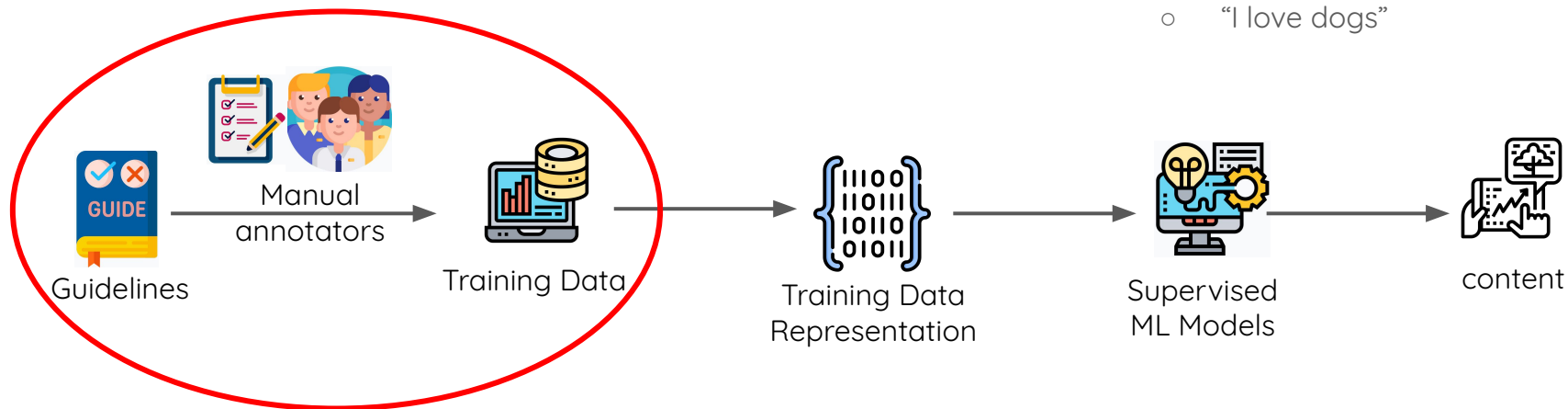


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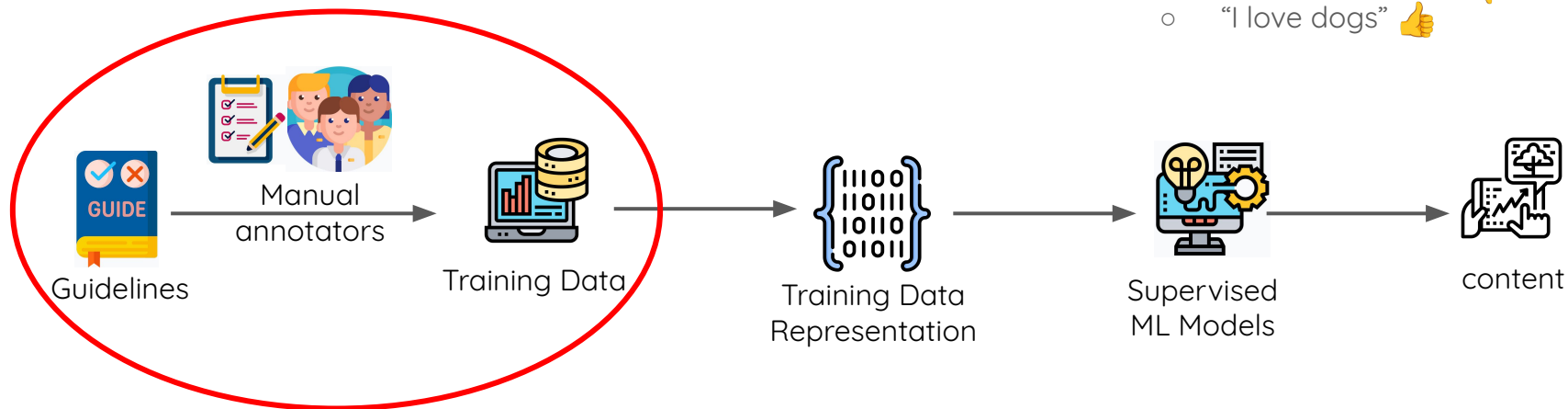


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- “I love cheese” 👍
- “Dogs hate cheese” 👎
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- “I love dogs” 👍

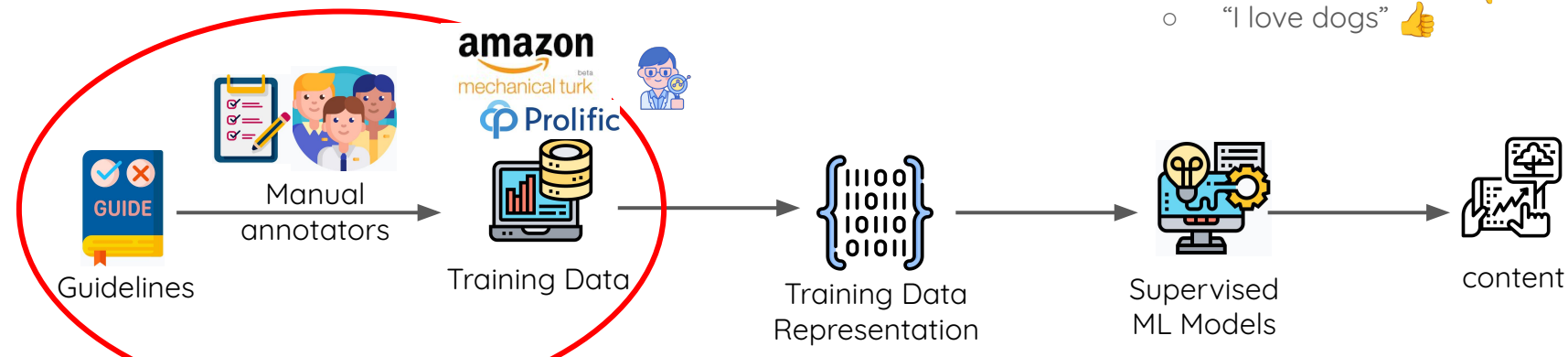


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Now, all we need is some **labeled data** or **training data**.

- “I love cheese” 👍
- “Dogs hate cheese” 👎
- “I hate chocolate” 👎
- “I love dogs” 👍



Usually labeled by humans, e.g., crowdworkers or research assistants

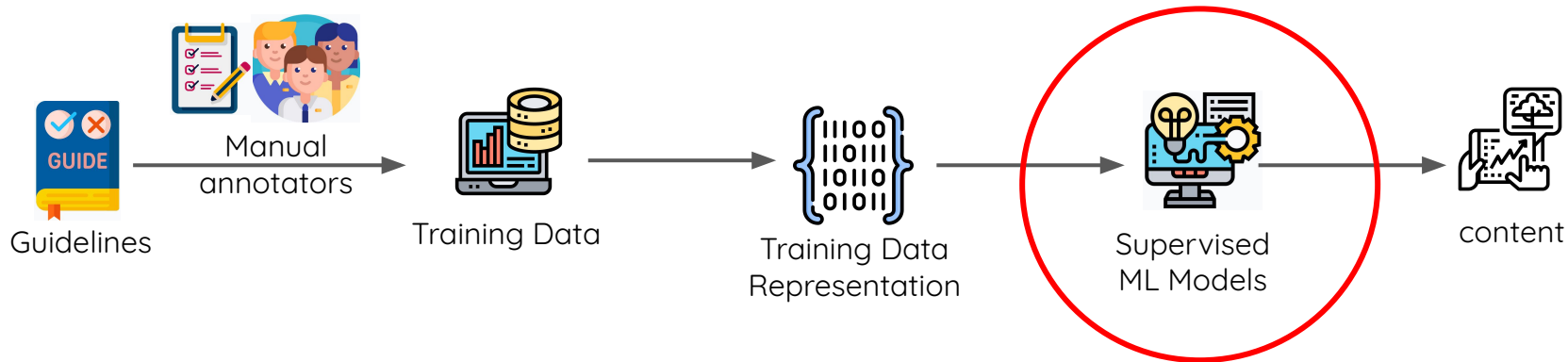
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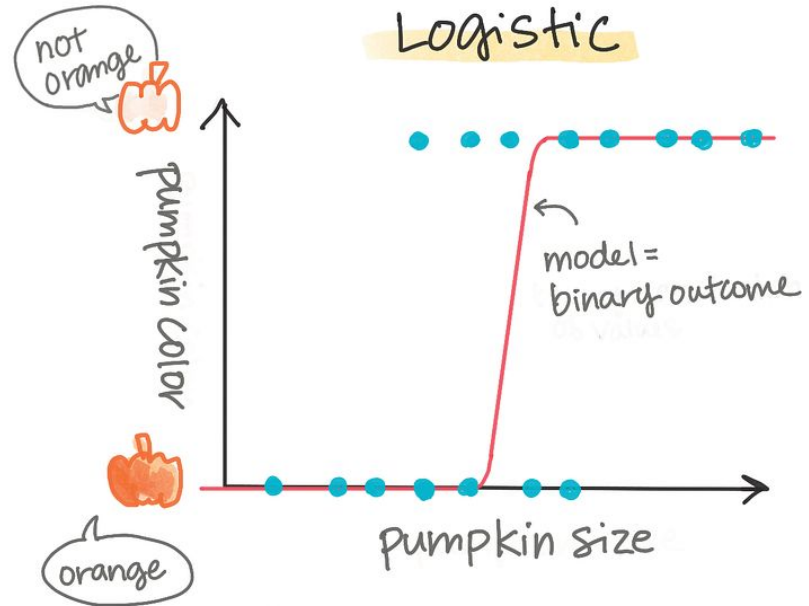
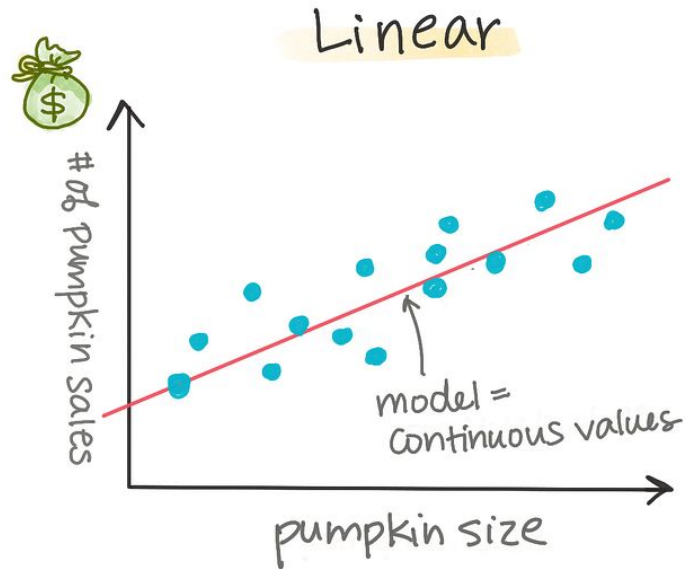
- “I love cheese” 👍
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And a **classifier**!



Classification: Logistic Regression

LINEAR vs. LOGISTIC REGRESSION

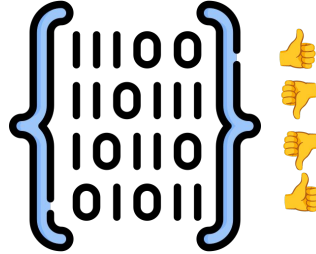


Classification: Logistic Regression

- “I love cheese” 👍
- “Dogs hate cheese” 👎
- “I hate chocolate” 👎
- “I love dogs” 👍

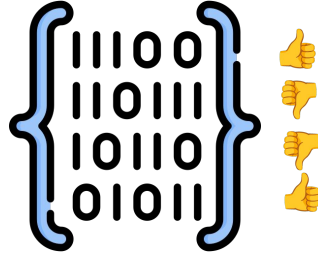
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Classification: Logistic Regression

- “I love cheese” 👍
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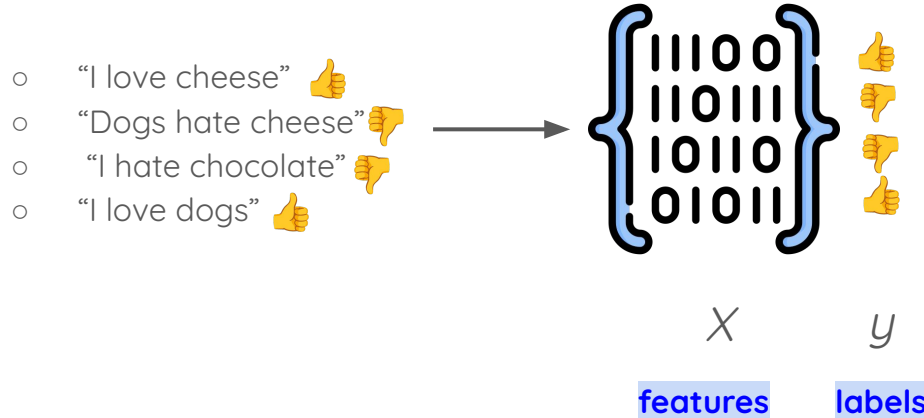
X

y

features

labels

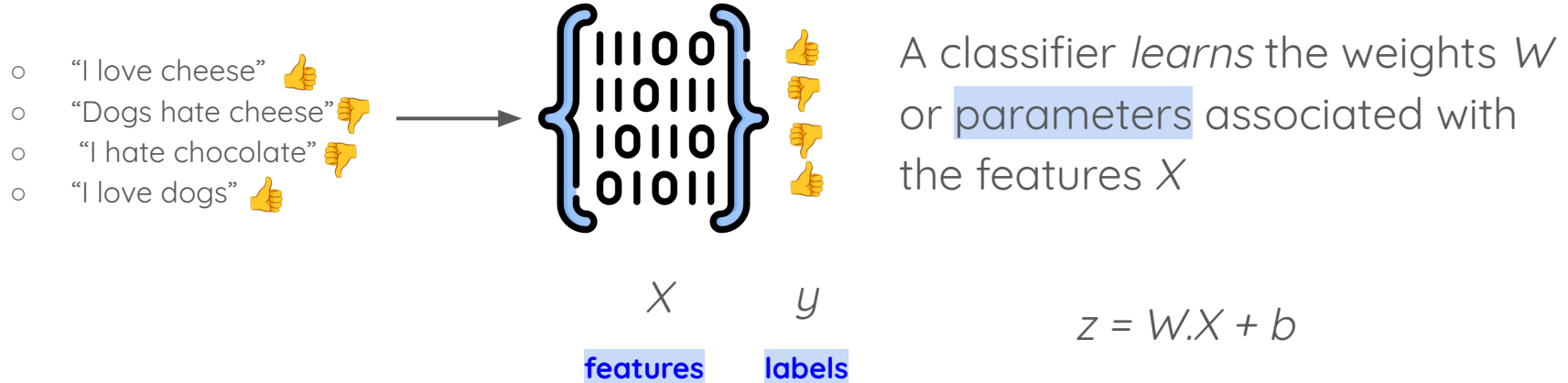
Classification: Logistic Regression



A classifier *learns* the weights W or parameters associated with the features X

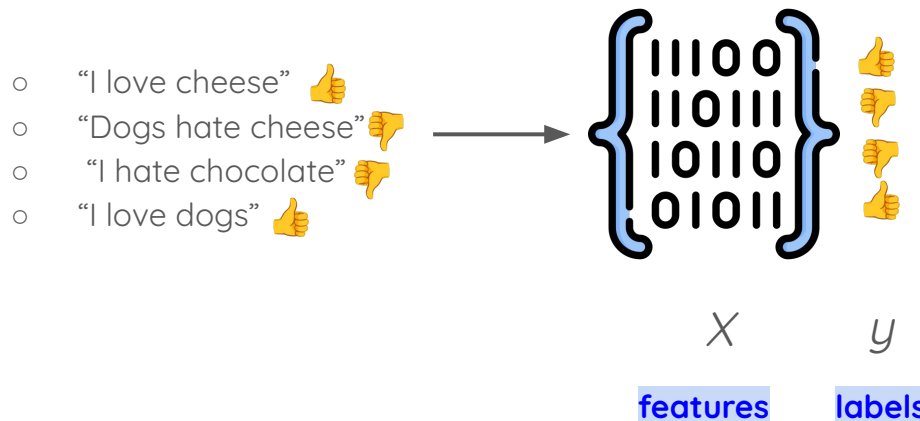
$$z = W.X + b$$

Classification: Logistic Regression



For each feature x_i , w_i tells us the importance of that feature We also have a bias term b .

Classification: Logistic Regression

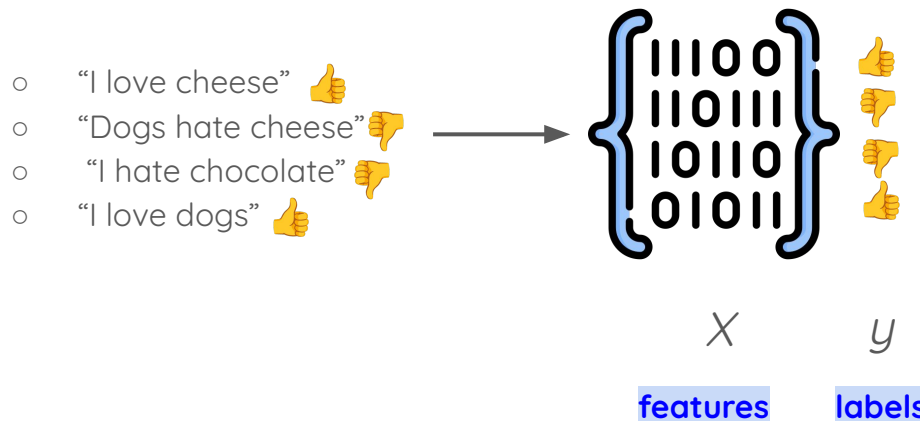


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For each feature x_i , w_i tells us the importance of that feature We also have a bias term b . But how do we get y (the labels) from z ?

Classification: Logistic Regression



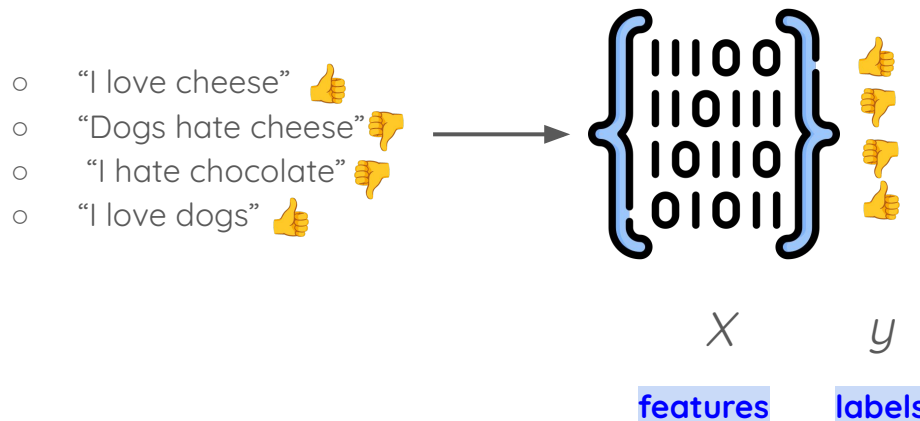
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Using the sigmoid or **logistic** function!

Classification: Logistic Regression



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For each feature x_i , w_i tells us the importance of that feature We also have a bias term b . But how do we get y (the labels) from z ?

Using the sigmoid or **logistic** function!

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

But how do we get w_i ? And what is $\hat{y}$?

$$z = W.X + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

But how do we get w_i ? And what is $\hat{y}$?

We can **estimate** y , i.e., $\hat{y}$

We start with some initial weights for W

Then **optimize** to W 's ideal value based on the difference between y and $\hat{y}$

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Concretely, we need:

$$z = W.X + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

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Then **optimize** to W 's ideal value based on the difference between y and $\hat{y}$

Concretely, we need:

- A **loss function** to measure the difference between y and $\hat{y}$

$$z = W.X + b$$

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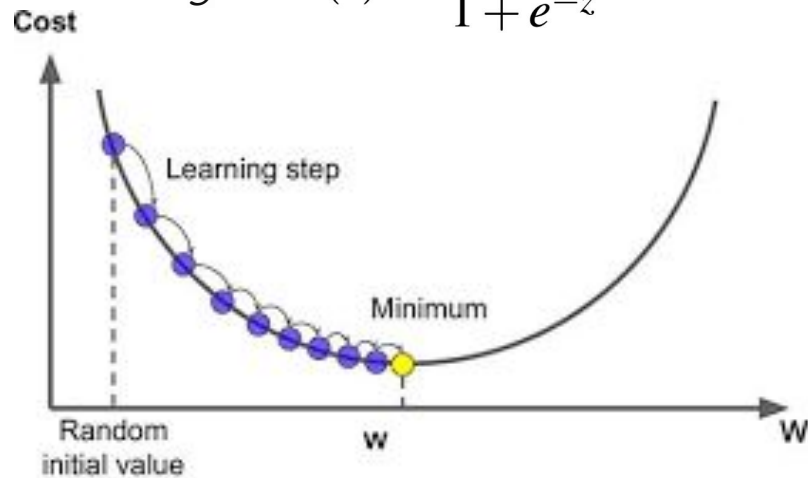
Then **optimize** to W 's ideal value based on the difference between y and $\hat{y}$

Concretely, we need:

- A **loss function** to measure the difference between y and $\hat{y}$
- An **optimization algorithm**

$$z = W.X + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$



Gradient Descent

Labeling unknown documents (also called **Inference** or **Prediction**)

For a new **unlabeled** document, we can get it's label using W

“I love chocolate”

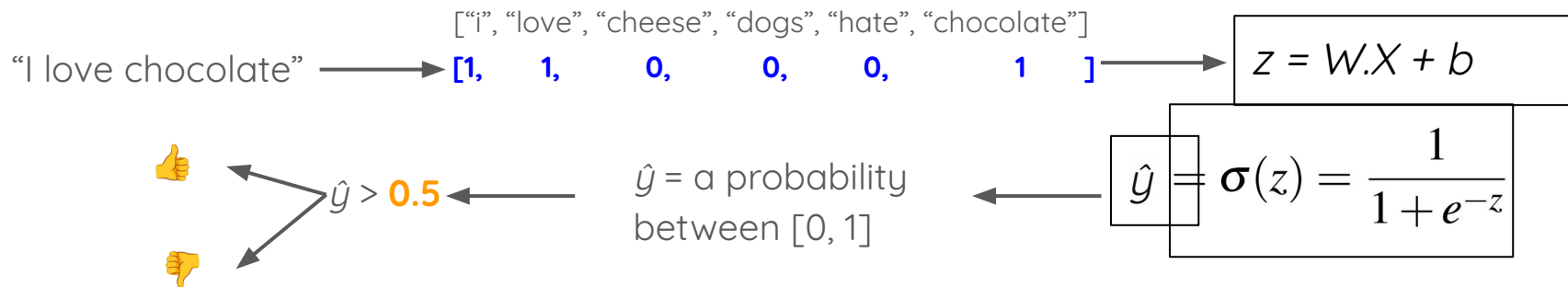
Labeling unknown documents (also called **Inference** or **Prediction**)

For a new **unlabeled** document, we can get its label using W and the new document's **feature representation**

“I love chocolate” $\longrightarrow$ $\begin{matrix} \text{[“I”, “love”, “cheese”, “dogs”, “hate”, “chocolate”]} \\ \text{[1, 1, 0, 0, 0, 1]} \end{matrix}$

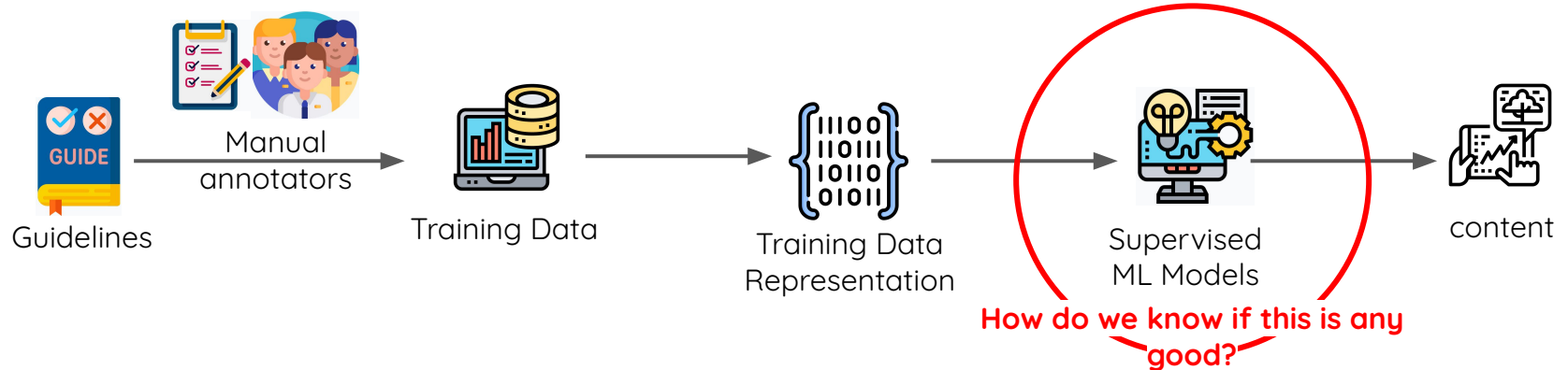
Labeling unknown documents (also called **Inference** or **Prediction**)

For a new **unlabeled** document, we can get its label using W and the new document's **feature representation** and a classification **threshold**:

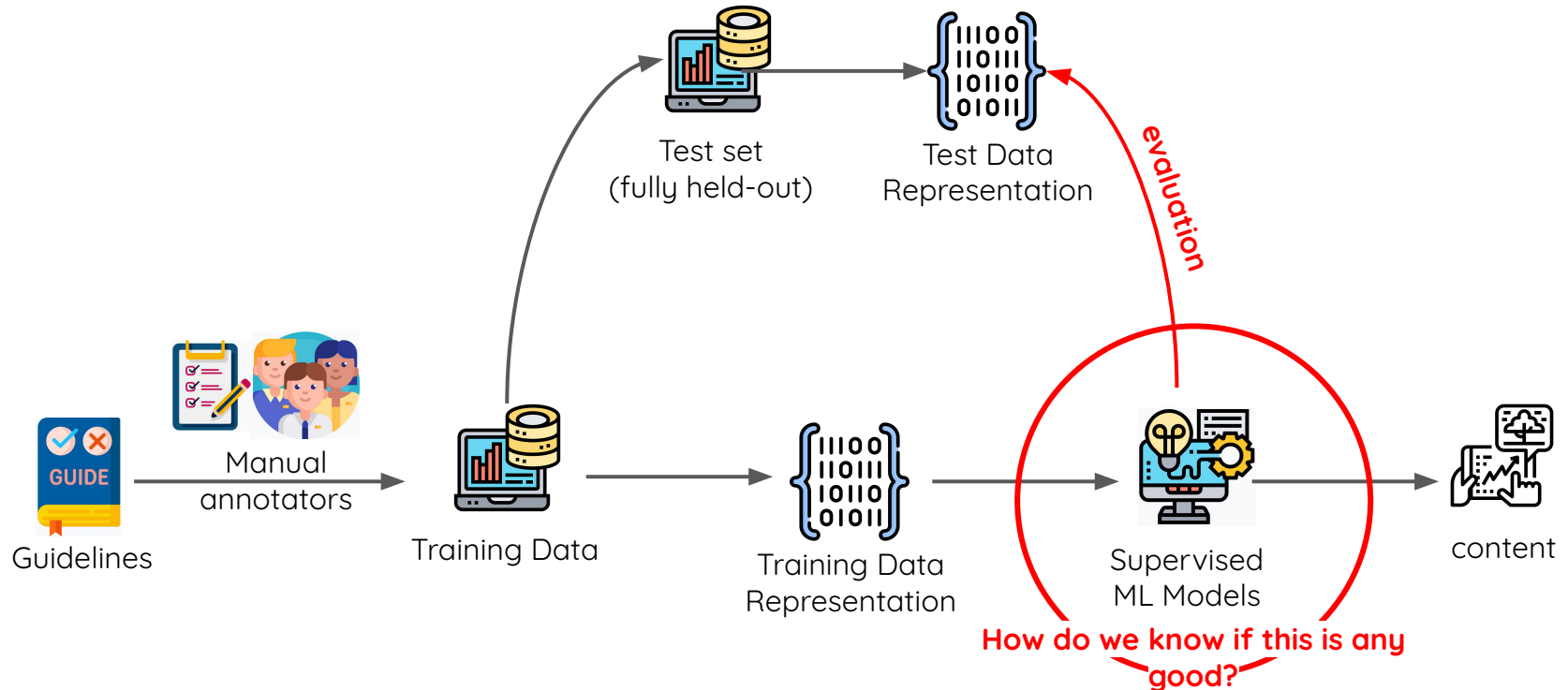


How do systematically evaluate the classifier's performance?

Evaluating Text Classification



Evaluating Text Classification



Evaluating Text Classification

Sources: [4][5][6][7][8][9][10][11][12] view · talk · edit

		Predicted condition			
		Positive (PP)	Negative (PN)	Informedness, bookmaker informedness (BM) $= \text{TPR} + \text{TNR} - 1$	Prevalence threshold (PT) $= \frac{\sqrt{\text{TPR} \times \text{FPR}} - \text{FPR}}{\text{TPR} - \text{FPR}}$
Actual condition	Total population $= P + N$				
	Positive (P)	True positive (TP), hit	False negative (FN), type II error, miss, underestimation	True positive rate (TPR), recall, sensitivity (SEN), probability of detection, hit rate, power $= \frac{\text{TP}}{P} = 1 - \text{FNR}$	False negative rate (FNR), miss rate $= \frac{\text{FN}}{P} = 1 - \text{TPR}$
	Negative (N)	False positive (FP), type I error, false alarm, overestimation	True negative (TN), correct rejection	False positive rate (FPR), probability of false alarm, fall-out $= \frac{\text{FP}}{N} = 1 - \text{TNR}$	True negative rate (TNR), specificity (SPC), selectivity $= \frac{\text{TN}}{N} = 1 - \text{FPR}$
	Prevalence $= \frac{P}{P + N}$	Positive predictive value (PPV), precision $= \frac{\text{TP}}{\text{PP}} = 1 - \text{FDR}$	False omission rate (FOR) $= \frac{\text{FN}}{\text{PN}} = 1 - \text{NPV}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$
	Accuracy (ACC) $= \frac{\text{TP} + \text{TN}}{P + N}$	False discovery rate (FDR) $= \frac{\text{FP}}{\text{PP}} = 1 - \text{PPV}$	Negative predictive value (NPV) $= \frac{\text{TN}}{\text{PN}}$ $= 1 - \text{FOR}$	Markedness (MK), deltaP (Δp) $= \text{PPV} + \text{NPV} - 1$	Diagnostic odds ratio (DOR) $= \frac{\text{LR}^+}{\text{LR}^-}$
	Balanced accuracy (BA) $= \frac{\text{TPR} + \text{TNR}}{2}$	F ₁ score $= \frac{2\text{PPV} \times \text{TPR}}{\text{PPV} + \text{TPR}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}$	Fowlkes–Mallows index (FM) $= \sqrt{\text{PPV} \times \text{TPR}}$	Matthews correlation coefficient (MCC) $= \frac{\sqrt{\text{TPR} \times \text{TNR} \times \text{PPV} \times \text{NPV}}}{\sqrt{\text{FNR} \times \text{FPR} \times \text{FOR} \times \text{FDR}}}$	Threat score (TS), critical success index (CSI), Jaccard index $= \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}$

Evaluating Text Classification

Sources: [4][5][6][7][8][9][10][11][12] view · talk · edit

		Predicted condition			
		Positive (PP)	Negative (PN)	Informedness, bookmaker informedness (BM) $= \text{TPR} + \text{TNR} - 1$	Prevalence threshold (PT) $= \frac{\sqrt{\text{TPR} \times \text{FPR}} - \text{FPR}}{\text{TPR} - \text{FPR}}$
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	Prevalence $= \frac{P}{P + N}$	Positive predictive value (PPV), precision $= \frac{\text{TP}}{\text{PP}} = 1 - \text{FDR}$	False omission rate (FOR) $= \frac{\text{FN}}{\text{PN}} = 1 - \text{NPV}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$
	Accuracy (ACC) $= \frac{\text{TP} + \text{TN}}{P + N}$	False discovery rate (FDR) $= \frac{\text{FP}}{\text{PP}} = 1 - \text{PPV}$	Negative predictive value (NPV) $= \frac{\text{TN}}{\text{PN}}$ $= 1 - \text{FOR}$	Markedness (MK), deltaP (Δp) $= \text{PPV} + \text{NPV} - 1$	Diagnostic odds ratio (DOR) $= \frac{\text{LR}^+}{\text{LR}^-}$
	Balanced accuracy (BA) $= \frac{\text{TPR} + \text{TNR}}{2}$	F ₁ score $= \frac{2\text{PPV} \times \text{TPR}}{\text{PPV} + \text{TPR}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}$	Fowlkes–Mallows index (FM) $= \sqrt{\text{PPV} \times \text{TPR}}$	Matthews correlation coefficient (MCC) $= \frac{\sqrt{\text{TPR} \times \text{TNR} \times \text{PPV} \times \text{NPV}}}{\sqrt{\text{FNR} \times \text{FPR} \times \text{FOR} \times \text{FDR}}}$	Threat score (TS), critical success index (CSI), Jaccard index $= \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}$

Evaluating Text Classification

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	Negative (N)	False positive (FP), type I error, false alarm, overestimation	True negative (TN), correct rejection	False positive rate (FPR), probability of false alarm, fall-out $= \frac{\text{FP}}{N} = 1 - \text{TNR}$	True negative rate (TNR), specificity (SPC), selectivity $= \frac{\text{TN}}{N} = 1 - \text{FPR}$
	Prevalence $= \frac{P}{P + N}$	Positive predictive value (PPV), precision $= \frac{\text{TP}}{\text{PP}} = 1 - \text{FDR}$	False omission rate (FOR) $= \frac{\text{FN}}{\text{PN}} = 1 - \text{NPV}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$
	Accuracy (ACC) $= \frac{\text{TP} + \text{TN}}{P + N}$	False discovery rate (FDR) $= \frac{\text{FP}}{\text{PP}} = 1 - \text{PPV}$	Negative predictive value (NPV) $= \frac{\text{TN}}{\text{PN}}$ $= 1 - \text{FOR}$	Markedness (MK), deltaP (Δp) $= \text{PPV} + \text{NPV} - 1$	Diagnostic odds ratio (DOR) $= \frac{\text{LR}^+}{\text{LR}^-}$
	Balanced accuracy (BA) $= \frac{\text{TPR} + \text{TNR}}{2}$	F ₁ score $= \frac{2\text{PPV} \times \text{TPR}}{\text{PPV} + \text{TPR}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}$	Fowlkes–Mallows index (FM) $= \sqrt{\text{PPV} \times \text{TPR}}$	Matthews correlation coefficient (MCC) $= \frac{\sqrt{\text{TPR} \times \text{TNR} \times \text{PPV} \times \text{NPV}}}{\sqrt{\text{FNR} \times \text{FPR} \times \text{FOR} \times \text{FDR}}}$	Threat score (TS), critical success index (CSI), Jaccard index $= \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}}$

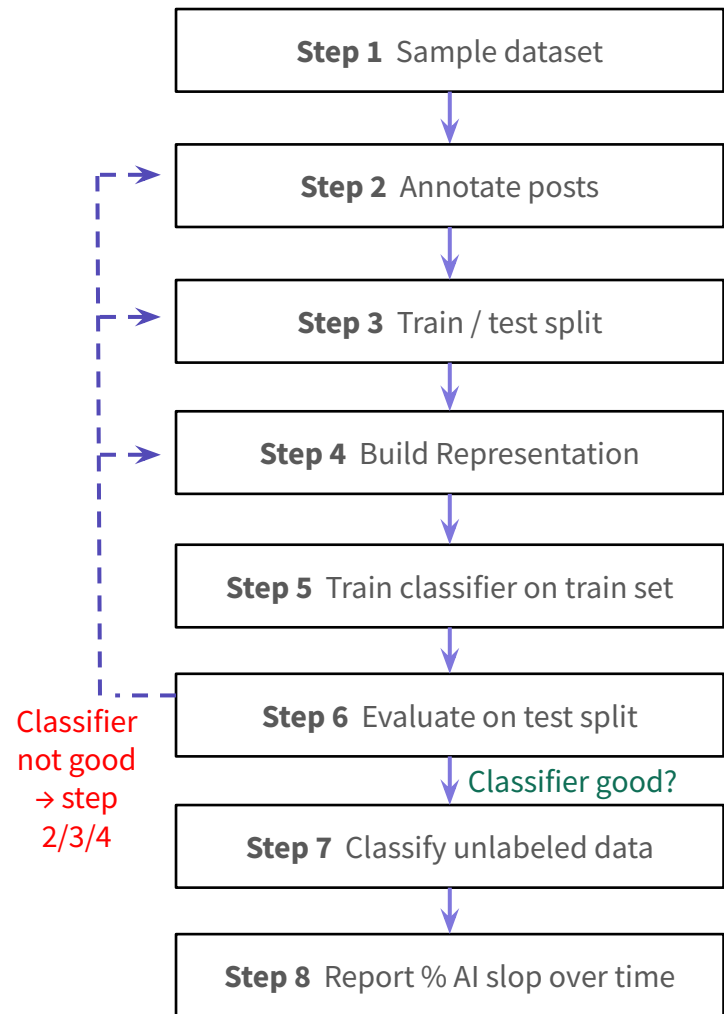
Check the worked out example in the appendix of this slide deck

Putting this all together for social media data analysis

- Let's say you want to study the 'AI-sloppiness' of some X/Twitter posts
- You don't have a readymade classifier for this
 - AI Sloppiness != AI-generated
- You have a cleaned dataset of X posts
- You need a **custom** classifier for your data and research question

Putting this all together for social media data analysis

- Let's say you want to study the 'AI-sloppiness' of some X/Twitter posts
- You don't have a readymade classifier for this
 - AI Sloppiness != AI-generated
- You have a cleaned dataset of X posts
- You need a **custom** classifier for your data and research question



Text **Classification** is not always straightforward

- Which and how many classes?
- What do the classes mean?
- Does a model do equally well on all classes?
- For student projects, getting labeled data is often a bottleneck
 - Depending on the classifier and task, you might need 200-1000 labeled examples (per class)
 - We will see how to circumvent this in the next lecture



Copyrighted Material
Bowker, Geoffrey C., and Susan Leigh Star.
[Sorting things out: Classification and its consequences](#). MIT press, 2000.

Summary and Next Week

- This session we learned:
 - Word Clouds, Word Shift Graphs, and other related visualizations
 - Different ways of analyzing social media text
 - Supervised Text Analysis: Sentiment Analysis with Logistic Regression
- Next time:
 - Text Analysis 2: more sophisticated classifiers and unsupervised ML
 - Readings:
 - Maria Antoniak's "Topic Modeling for the People":
<https://maria-antoniak.github.io/2022/07/27/topic-modeling-for-the-people.html>
 - Mitra, Tanushree, and Eric Gilbert. "[The language that gets people to give: Phrases that predict success on kickstarter.](#)"
Proceedings of the 17th ACM conference on Computer supported cooperative work & social computing. 2014.
- Please register your teams: [Project Registration IS 618 Social Media Data Analysis](#)
 - Deadline: 16.04
 - Project pitches during the lecture slot on 21.04
 - Project presentations will be done via my laptop. Upload slides to ILIAS by 21.04 11 AM

Questions?

Additional Resources

- Uni Mannheim course on Text Analytics:
<https://www.bwl.uni-mannheim.de/modulkatalog-mmm/ie-661-text-analytics/>
- Online Text Classification course from Lene Voita:
https://lena-voita.github.io/nlp_course/text_classification.html#main_content

Appendix: Worked out Example for Measuring a Classifier's Accuracy

Confusion Matrix: Basic, but useful



Actual Values

	Positive (1)	Negative (0)
 Predicted Values		
Positive (1)	TP	FP
Negative (0)	FN	TN

Example for Sentiment Analysis

- “Ground truth” or “gold labels” = manual labels



Example for Sentiment Analysis

- “Ground truth” or “gold labels” = manual labels



- True positive (TP) = when something that was manually labeled as positive was also labeled positive by the classifier



“I had a **great** day today”



Example for Sentiment Analysis

- “Ground truth” or “gold labels” = manual labels



- True positive (TP) = when something that was manually labeled as positive was also labeled positive by the classifier



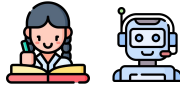
“I had a **great** day today”



- True negative (TN) = when something that was manually labeled as negative was also labeled negative by the classifier



“I had an **awful** day today”



Example for Sentiment Analysis

- “Ground truth” or “gold labels” = manual labels



- True positive (TP) = when something that was manually labeled as positive was also labeled positive by the classifier



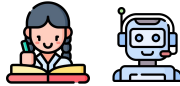
“I had a **great** day today”



- True negative (TN) = when something that was manually labeled as negative was also labeled negative by the classifier



“I had an **awful** day today”



- False positive (FP) = when something that was manually labeled as negative was **falsely** labeled positive by the classifier



“I had a **not so great** day today”



Example for Sentiment Analysis

- “Ground truth” or “gold labels” = manual labels



- True positive (TP) = when something that was manually labeled as positive was also labeled positive by the classifier



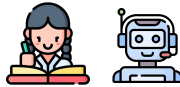
“I had a **great** day today”



- True negative (TN) = when something that was manually labeled as negative was also labeled negative by the classifier



“I had an **awful** day today”



- False positive (FP) = when something that was manually labeled as negative was **falsely** labeled positive by the classifier



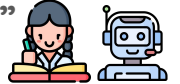
“I had a **not so great** day today”



- False negative (FN) = when something that was manually labeled as positive was **falsely** labeled negative by the classifier



“I had an **awfully great** day today”



Example for Sentiment Analysis

- Let's say we have 100 item in our test set, i.e., 100 social media posts manually annotated into positive (**56**) and negative (**44**), i.e., $N_{\text{classes}} = 2$
- The classifier labels
 - __ posts as positive
 - __ posts as negative

	True +ve	True -ve
Predicted +ve		
Predicted -ve		
	56	44

Example for Sentiment Analysis

- Let's say we have 100 item in our test set, i.e., 100 social media posts manually annotated into positive (**56**) and negative (**44**), i.e., $N_{\text{classes}} = 2$
- The classifier labels
 - __ posts as positive
 - __ posts as negative

	True +ve	True -ve
Predicted +ve	48	3
Predicted -ve	8	41
	56	44

Example for Sentiment Analysis

- Let's say we have 100 item in our test set, i.e., 100 social media posts manually annotated into positive (**56**) and negative (**44**), i.e., $N_{\text{classes}} = 2$
- The classifier labels
 - 51 posts as positive
 - 49 posts as negative

	True +ve	True -ve	
Predicted +ve	48	3	51
Predicted -ve	8	41	49
	56	44	

Example for Sentiment Analysis

- Let's say we have 100 item in our test set, i.e., 100 social media posts manually annotated into positive (**56**) and negative (**44**), i.e., $N_{\text{classes}} = 2$
- The classifier labels
 - 51 posts as positive
 - 49 posts as negative
- TPs:
- TNs:
- FPs:
- FNs:

	True +ve	True -ve	
Predicted +ve	48	3	51
Predicted -ve	8	41	49
	56	44	

Example for Sentiment Analysis

- Let's say we have 100 item in our test set, i.e., 100 social media posts manually annotated into positive (**56**) and negative (**44**), i.e., $N_{\text{classes}} = 2$
- The classifier labels
 - 51 posts as positive
 - 49 posts as negative
- TPs: 48
- TNs:
- FPs:
- FNs:

	True +ve	True -ve	
Predicted +ve	48	3	51
Predicted -ve	8	41	49
	56	44	

Example for Sentiment Analysis

- Let's say we have 100 item in our test set, i.e., 100 social media posts manually annotated into positive (**56**) and negative (**44**), i.e., $N_{\text{classes}} = 2$
- The classifier labels
 - 51 posts as positive
 - 49 posts as negative
- TPs: 48
- TNs: 41
- FPs:
- FNs:

	True +ve	True -ve	
Predicted +ve	48	3	51
Predicted -ve	8	41	49
	56	44	

Example for Sentiment Analysis

- Let's say we have 100 item in our test set, i.e., 100 social media posts manually annotated into positive (**56**) and negative (**44**), i.e., $N_{\text{classes}} = 2$
- The classifier labels
 - 51 posts as positive
 - 49 posts as negative
- TPs: 48
- TNs: 41
- FPS: 3
- FNs: 8

	True +ve	True -ve	
Predicted +ve	48	3	51
Predicted -ve	8	41	49
	56	44	

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- Let's say we have 100 item in our test set, i.e., 100 social media posts manually annotated into positive (**56**) and negative (**44**), i.e., $N_{\text{classes}} = 2$

- The classifier labels
 - 51 posts as positive
 - 49 posts as negative

- TPs: 48
- TNs: 41
- FPS: 3
- FNs: 8

Accuracy (ACC)

$$= \frac{TP + TN}{P + N}$$

- Accuracy =

	True +ve (P)	True -ve (N)	
Predicted +ve	48	3	51
Predicted -ve	8	41	49
	56	44	

Example for Sentiment Analysis

- Let's say we have 100 item in our test set, i.e., 100 social media posts manually annotated into positive (**56**) and negative (**44**), i.e., $N_{\text{classes}} = 2$

- The classifier labels
 - 51 posts as positive
 - 49 posts as negative

- TPs: 48
- TNs: 41
- FPS: 3
- FNs: 8

Accuracy (ACC)

$$= \frac{TP + TN}{P + N}$$

- Accuracy = $(48 + 41)/(100) = 89\%$

	True +ve (P)	True -ve (N)	
Predicted +ve	48	3	51
Predicted -ve	8	41	49
	56	44	

Example for Sentiment Analysis: Imbalanced classes

- Let's say we have 100 item in our test set, i.e., 100 social media posts manually annotated into positive (**96**) and negative (**4**), i.e., $N_{\text{classes}} = 2$
- The classifier labels
 - 99 posts as positive
 - 1 post as negative

- TPs:
- TNs:
- FPs:
- FNs:

Accuracy (ACC)

$$= \frac{TP + TN}{P + N}$$

- Accuracy =

	True +ve (P)	True -ve (N)	
Predicted +ve	96	3	99
Predicted -ve	0	1	1
	96	4	

Example for Sentiment Analysis: Imbalanced classes

- Let's say we have 100 item in our test set, i.e., 100 social media posts manually annotated into positive (**96**) and negative (**4**), i.e., $N_{\text{classes}} = 2$
- The classifier labels
 - 99 posts as positive
 - 1 post as negative

- TPs: 96
- TNs: 1
- FPs: 3
- FNs: 0

Accuracy (ACC)

$$= \frac{TP + TN}{P + N}$$

- Accuracy = $(96 + 1)/(100) = 97\%$

	True +ve (P)	True -ve (N)	
Predicted +ve	96	3	99
Predicted -ve	0	1	1
	96	4	